Score: 157 / 175

ID: 22402348
PTS: 1

Defines the services that are provided by the component to other components.

- **Provides interface**
- Requires interface
- Emergent Property Prediction

ID: 22402349
PTS: 1

The component model specifies how the interfaces should be defined and the elements, such as operation names, parameters and exceptions, which should be included in the interface definition.

- Usage
- Deployment
- **Interfaces**

ID: 22402351
PTS: 1

The process of acquiring components for reuse or development into a reusable component.

- Component management
- Component certification
- **Component acquisition**

ID: 22402352
PTS: 1

Components developed for a specific application usually have to be generalised to make them reusable.

- False
- **True**

ID: 22402353
PTS: 0

This involves writing a wrapper component that implements provide and not requires interfaces then accesses the legacy system.

- False
- **True**

ID: 22402354
PTS: 1

Generic components may be less space-

efficient and may have no longer execution times than their specific equivalents.

- **False**
- True

ID: 22402355
PTS: 1

When reusing components, it is essential to make trade-offs between ideal requirements and the services actually provided by available components.

- **True**
- False

ID: 22402356
PTS: 1

That provides support for component inter-operability.

- Independent components
- Component standards
- **Middleware**

ID: 22402357
PTS: 0

Composition involves integrating components with each other and with the

component infrastructure.

- **True**
- False

ID: 22402358
PTS: 1

It is where an attacker generates information that should not exist and then uses this to gain some privileges.

- Modification
- **Fabrication**
- Interception

ID: 22402359
PTS: 1

Involves the sending computer sending information about what is required to another computer.

- **Message-based interaction**
- Failure management
- Procedural interaction

ID: 22402360
PTS: 1

Use of equipment and software from different vendors.

- **Openness**
- Resource sharing
- Concurrency

ID: 22402361
PTS: 1

More processing is delegated to the client as the application processing is locally executed.

- **True**
- False

ID: 22402362
PTS: 1

Users may pay for the software according to the amount of use they make of it or through an annual or monthly subscription.

- False
- **True**

ID: 22402363
PTS: 1

Users may not pay for the software according to the amount of use they make

of it or through an annual or monthly subscription.

- True
- **False**

ID: 22402365
PTS: 1

Service-oriented architecture is an approach to structuring a software system as a set of separate, stateless services.

- False
- **True**

ID: 22402367
PTS: 1

The ability to continue in operation after a fault has occurred.

- Resource sharing
- **Fault Tolerance**
- Openness

ID: 22402372
PTS: 1

A unit of composition with contractually specified interfaces and explicit context dependencies only.

- Councill and Heinmann
- **Szyperski**
- Emergent Property Prediction

ID: 22402373
PTS: 1

Defines the services that specifies what services must be made available for the component to execute as specified.

- Emergent Property Prediction
- Provides interface
- **Requires interface**

ID: 22402374
PTS: 1

The operation of the component may depend on local exception handling, and changing this may have serious implications for the functionality of the component.

- False
- **True**

ID: 22402375
PTS: 1

Generic components may be less space-

efficient and may have longer execution times than their specific equivalents.

- False
- **True**

ID: 22402376
PTS: 1

Different types of adaptor are required depending on the type of composition.

- **True**
- False

ID: 22402377
PTS: 1

Different types of adaptor are not required depending on the type of composition.

- True
- **False**

ID: 22402378
PTS: 1

The emergent properties of component compositions be predicted

- **Emergent Property**

Prediction
- Component Certification
- Component trustworthiness

ID: 22402379
PTS: 1

Components developed for a specific application usually have to be generalised to make them reusable.

- False
- **True**

ID: 22402382
PTS: 1

Concurrent processing to enhance performance.

- Resource sharing
- Openness
- **Concurrency**

ID: 22402383
PTS: 1

The overall system is not designed to take advantage of the computational power and storage of a large number of networked computers.

- **False**
- True

ID: 22402384
PTS: 1

Software is deployed on a server (or more commonly a number of servers) and is accessed through a web browser.

- **True**
- False

ID: 22402385
PTS: 1

The server maintains the user's data and state during an interaction session.

- **True**
- False

ID: 22402387
PTS: 1

Branding, where users from each organization, are not presented with an interface that reflects their own organization.

- **False**

- True

ID: 22402392
PTS: 1

Specifies how interfaces should be defined and the elements that should be included in an interface definition.

- service model
- **component model**
- distributed model

ID: 22402393
PTS: 1

It concern with managing a company's reusable components, ensuring that they are properly catalogued, stored and made available for reuse.

- Component certification
- **Component management**
- Component acquisition

ID: 22402394
PTS: 0

Components developed for a specific application usually have to be generalised to make them usable.

- **True**
- False

ID: 22402395
PTS: 0

Components for use may be specially constructed by generalising existing components.

- False
- **True**

ID: 22402397
PTS: 1

The development cost of reusable components may be higher than the cost of specific equivalents.

- **True**
- False

ID: 22402398
PTS: 1

The component specification may not be detailed enough to allow comprehensive tests to be developed.

- **True**
- False

ID: 22402399
PTS: 1

Where data or services in the system are changed by an attacker.

- Interruption
- **Modification**
- Interception

ID: 22402401
PTS: 1

Software as a service is a way of providing functionality on a remote server with client access through a web browser.

- False
- **True**

ID: 22402402
PTS: 1

To what extent should the distributed system appear to the user as a single system?

- Openness
- Scalability

- **Transparency**

ID: 22402404
PTS: 1

In order for components to be distributed and accessed remotely, they need to have a unique name or handle associated with them.

- Deployment
- **Usage**
- Interface

ID: 22402405
PTS: 1

The component specification maybe detailed enough to allow comprehensive tests to be developed.

- True
- **False**

ID: 22402406
PTS: 1

The major problem with component validation is that the component specification maybe sufficiently detailed to

allow you to develop a complete set of component tests.

- True
- **False**

ID: 22402407
PTS: 1

The functionality that failed in this component was required in Ariane 5.

- **False**
- True

ID: 22402408
PTS: 1

The Object Constraint Language (OCL) has been designed to define constraints that are associated with UML models.

- False
- **True**

ID: 22402409
PTS: 1

It is where system services are attacked and cannot be delivered as expected.

- Modification
- **Interruption**
- Interception

ID: 22402411
PTS: 1

More complex than a fat client model especially for management.

- **False**
- True

ID: 22402412
PTS: 1

More processing is delegated to the client as the application processing is locally not executed.

- **False**
- True

ID: 22402413
PTS: 1

Certification means that someone apart from the developer checks the quality of the component.

- **True**
- False

ID: 22402414
PTS: 1

The functionality that failed in this component was not required in Ariane 5.

- False
- **True**

ID: 22402415
PTS: 1

This is will certify the quality of components

- **Component Certification**
- Component trustworthiness
- Component standards

ID: 22402416
PTS: 1

It should be possible to manage a system as it increases in size, even if parts of the system are located in independent organizations.

- Size
- Distribution

- **Manageability**

ID: 22402417
PTS: 1

It is where some or all of the application processing is carried out on the client.

- **Fat-client model**
- Failure management
- Thin-client model

ID: 22402418
PTS: 0

This avoids problems with scalability and performance if a fat-client two-tier model is chosen, or problems of system management if a fat-client model is used.

- False
- **True**

ID: 22402419
PTS: 1

Software is not deployed on a server (or more commonly a number of servers) and is accessed through a web browser.

- True
- **False**

ID: 22402432
PTS: 1

This is to facilitate component integration.

- Independent components
- Middleware
- **Component standards**

ID: 22402433
PTS: 0

When using components, it is essential to make trade-offs between ideal requirements and the services actually provided by available components.

- **True**
- False

ID: 22402434
PTS: 1

It can a component with no available source code be trusted.

- Component Certification
- Component standards

- **Component trustworthiness**

ID: 22402435
PTS: 1

Increased throughput by adding new resources.

- Resource sharing
- Openness
- **Scalability**

ID: 22402436
PTS: 0

Service-oriented architecture is not an approach to structuring a software system as a set of separate, stateless services.

- **False**
- True

ID: 22402437
PTS: 1

It should be possible to add more resources to a system to cope with increasing numbers of users.

- Manageability
- Distribution
- **Size**

ID: 22402438
PTS: 1

Components for reuse may be specially constructed by generalising existing components.

- False
- **True**

ID: 22402439
PTS: 1

Composition involves integrating components with each other and with the component infrastructure.

- False
- **True**

ID: 22402440
PTS: 1

What is Quality of Service?

- should include mechanisms for discovering if a component of the

system has failed.

- it is inevitable that failures will occur, so the system has to be designed to be resilient to these failures.
- **offered by a distributed system reflects the system's ability to deliver its services dependably and with a response time and throughput that is acceptable to its users.**

ID: 22402441
PTS: 1

There is no distinction in a distributed component architecture between clients and servers.

- False
- **True**

ID: 22402442
PTS: 1

The server no maintains the user's data and state during an interaction session.

- **False**
- True

ID: 22402443
PTS: 1

Access control, where service customers create individual accounts for their staff and define the resources and functions that are accessible to each of their users.

- False
- **True**

ID: 22402466
PTS: 1

What is Component-based development?

- components developed using different approaches to work together.
- provides a service without regard to where the component is executing or its programming language
- **. is an approach to software development that relies on the reuse of entities called 'software components.**

ID: 22402468
PTS: 1

In 1996, the 1st test flight of the Ariane 5 rocket ended in disaster when the launcher went out of control 47 seconds after take off.

- True
- **False**

ID: 22402469
PTS: 1

More complex than a thin client model especially for management.

- False
- **True**

ID: 22402470
PTS: 1

Can system failures be detected, contained and repaired

- **Failure management**
- Scalability
- Security

ID: 22413250
PTS: 1

Where communications between parts of the system are intercepted by an attacker so that there is a loss of confidentiality

- Modification
- Interruption

- **Interception**

ID: 22413251
PTS: 1

A major advantage is that it places a heavy processing load on both the server and the network.

- **False**
- True

ID: 22413252
PTS: 1

There is distinction in a distributed component architecture between clients and servers.

- **False**
- True

ID: 22413590
PTS: 1

The process of checking a component and certifying that it meets its specification.

- **Component certification**
- Component management
- Component acquisition

ID: 22413591
PTS: 1

Certification means that someone apart from the developer checks the inequality of the component.

- **False**
- True

ID: 22413592
PTS: 0

This is geared to reuse.

- Development Process
- Component standards
- **Independent components**

ID: 22413593
PTS: 1

The Object Constraint Language (OCL) has been not designed to define constraints that are associated with UML models.

- **False**
- True

ID: 22416034
PTS: 0

This involves writing a wrapper component that implements provide and requires interfaces then accesses the legacy system.

- **False**
- True

ID: 22466358
PTS: 1

A message exchange standard that supports service communication

- WS-BPEL
- WSDL (Web Service Definition Language)
- **SOAP**

ID: 22466359
PTS: 1

Your selection criteria will obviously include the functionality of the services offered. They may also include the cost of the services and the quality of service (responsiveness, availability, etc.) offered.

- Discover services
- **Select possible services**
- Formulate outline workflow

ID: 22466360
PTS: 1

Create workflow program during this stage, the abstract workflow design is not-transformed to an executable program and the service interface is defined.

- **False**

- True

ID: 22466361
PTS: 1

Test completed service or application the process of testing the completed, composite service is more complex than component testing in situations where external services are used.

- False
- **True**

ID: 22466363
PTS: 1

Graphical workflow notations, such as BPMN, are more readable and WS-BPEL cannot be generated from them.

- **False**
- True

ID: 22466364
PTS: 1

Testing is intended to find defects and demonstrate that a system meets its functional and functional requirements.

- True

- **False**

ID: 22466366
PTS: 1

It may be easy to invoke compensating actions in external services as these may rely on the failure of other services which cannot be simulated.

- **False**
- True

ID: 22466368
PTS: 1

Which has a shorter term outlook, and focuses on planning the next increment of a system.

- **Iteration Planning**
- Validation Planning
- Release Planning

ID: 22466369
PTS: 1

It is where code is not modified an effort estimate (PM) is computed.

- White-box
- Green-box

- **Black-box**

ID: 22466370
PTS: 0

PM = (ASLEC * AT/100)/ATPROD.

- False
- **True**

ID: 22466371
PTS: 1

ATPROD is the productivity of engineers in integrating this code.

- **True**
- False

ID: 22466372
PTS: 1

ESLOC = ASLOC * (1-AT/100) * AAM.

- False
- **True**

ID: 22466373
PTS: 1

This depends on 5 scale factors (see next

slide). Their sum/100 is added to 1.012.

- **False**
- True

ID: 22466374
PTS: 1

Product attributes Concerned with required characteristics of the software product being developed.

- False
- **True**

ID: 22466375
PTS: 1

Describes the quality procedures and standards that will be used in a project.

- Validation plan
- **Quality plan**
- Configuration management plan

ID: 22466376
PTS: 1

Personnel attributes Multipliers that take the experience and capabilities of the people not working on the project into account.

- **False**
- True

ID: 22466377
PTS: 1

The time required is independent of the number of people working on the project.

- False
- **True**

ID: 22466387
PTS: 0

What is Web Services?

- **Standard protocols have been developed to support service communication and information exchange**
- may execute on different computers from different service providers
- an instance of a more general notion of a service

ID: 22466389
PTS: 1

Design the structure and organisation of the input and output messages. Notations such as the UML are a more abstract

representation than XML

- **Message design**
- Logical interface design
- WSDL description

ID: 22466390
PTS: 1

In inter-organisational systems, separate workflows are created for each organisation and linked through message exchange.

- **True**
- False

ID: 22466391
PTS: 1

WS-BPEL is an XML-standard for over workflow specification.

- **False**
- True

ID: 22466392
PTS: 1

Dynamic binding means that the service used in an application may vary - the application tests are not, therefore,

unreliable.

- True
- **False**

ID: 22466393
PTS: 1

If services have to be paid for as used, testing a service may be expensive.

- **True**
- False

ID: 22466394
PTS: 1

If services have to be paid for as used, testing a service may be not expensive.

- True
- **False**

ID: 22466395
PTS: 1

Coordination Services that not support composite processes such as ordering.

- **False**
- True

ID: 22466398
PTS: 1

Used when software is composed from existing parts.

- **Application composition model**
- Reuse model
- Early design model

ID: 22466399
PTS: 1

Used to compute the effort of integrating reusable components.

- Application composition model
- **Reuse model**
- Early design model

ID: 22466400
PTS: 1

This depends on 5 scale factors (see next slide). Their sum/100 is added to 1.01.

- **True**
- False

ID: 22466401
PTS: 1

Computer attributes Constraints imposed on the software by the hardware platform.

- **True**
- False

ID: 22466402
PTS: 1

Project attributes Concerned with the particular characteristics of the hardware development project.

- True
- **False**

ID: 22466403
PTS: 1

Describes the approach, resources, and schedule used for system validation.

- **Validation plan**
- Configuration management plan
- Quality plan

ID: 22466404
PTS: 1

This are normally used to illustrate the project schedule.

- bar chart
- validation plan
- **graphical Notation**

ID: 22466418
PTS: 1

It support composite processes such as ordering.

- Utility Services
- Business Services
- **Coordination Services**

ID: 22466420
PTS: 1

Starts with the service requirements and defines the operation names and parameters associated with the service.

- Message design
- WSDL description
- **Logical interface design**

ID: 22466421
PTS: 1

External services may not have modified by the service provider thus invalidating tests which have been completed.

- **False**
- True

ID: 22466422
PTS: 1

A standard for workflow languages used to define service composition

- SOAP
- **WS-BPEL**
- WSDL (Web Service Definition Language)

ID: 22466423
PTS: 1

The non-functional behavior of the service is predictable because it depends on load.

- True
- **False**

ID: 22466424
PTS: 1

What is Project Planning?

- **involves breaking down the work into parts and assign these to project team members, anticipate problems that might arise and prepare tentative solutions to those problems.**
- has to plan who will work on the project, how the project will be broken down into increments, how resources will be allocated across your company, etc.
- when you modify your plan in the light of experience gained and information from monitoring the progress of the work.

ID: 22466425
PTS: 1

It looks ahead for several months and decides on the features that should be included in a release of a system.

- Validation Planning
- **Release Planning**
- Iteration Planning

ID: 22466427
PTS: 0

It is where code is modified a size estimate equivalent to the number of lines of new source code is computed.

- **Green-box**
- Black-box
- White-box

ID: 22466428
PTS: 1

EAM is the adaptation adjustment multiplier computed from the costs of changing the reused code.

- True
- **False**

ID: 22466429
PTS: 1

A company takes on a project in a new

domain. The client has not defined the process to be used and has not allowed time for risk analysis. The company has a CMM level 2 rating.

- **True**
- False

ID: 22466430
PTS: 1

Product attributes Concerned with required characteristics of the software product being not developed.

- **False**
- True

ID: 22467480
PTS: 1

The overall system is designed to take advantage of the computational power and storage of a large number of networked computers.

- False
- **True**

ID: 22469296
PTS: 1

This is associated with a specific business function e.g., in a university, student registration.

- Utility Services
- Coordination Services
- **Business Services**

ID: 22469297
PTS: 1

In this initial stage of service design, you use the requirements for the composite service as a basis for creating an 'ideal' service design.

- Select possible services
- **Formulate outline workflow**
- Discover services

ID: 22469298
PTS: 1

Test completed service or application the process of testing the completed, composite service is more complex than component testing in situations where internal services are used.

- **False**
- True

ID: 22469299
PTS: 1

It is where you design the logical and WSDL service interfaces.

- Service implementation and deployment
- **Service design**
- Service candidate identification

ID: 22469300
PTS: 1

That implement general functionality used by different business processes.

- **Utility Services**
- Business Services
- Coordination Services

ID: 22469301
PTS: 1

AT is the percentage of code automatically generated.

- **True**
- False

ID: 22469302
PTS: 1

AAM is the adaptation adjustment multiplier computed from the costs of changing the reused code.

- False
- **True**

ID: 22469303
PTS: 1

Computer attributes Constraints imposed on the software by the software platform.

- **False**
- True

ID: 22469304
PTS: 1

PM is the effort computation and B is the exponent computed as discussed above (B is 1 for the early prototyping model).

- False
- **True**

ID: 22469305
PTS: 1

During this stage of the process, you

search service registries or catalogs to discover what services exist, who provides these services and the details of the service provision.

- Select possible services
- Formulate outline workflow
- **Discover services**

ID: 22469306
PTS: 1

Refine workflow this involves adding detail to the abstract description and perhaps adding or removing workflow activities.

- False
- **True**

ID: 22469307
PTS: 1

Refine workflow this involves adding detail to the abstract description and perhaps not adding or removing workflow activities.

- **False**
- True

ID: 22469309
PTS: 1

It is where you implement and test the service and make it available for use.

- Service candidate identification
- **Service implementation and deployment**
- Service design

ID: 22469310
PTS: 1

ASLOC is the not number of lines of generated code.

- **False**
- True

ID: 22469312
PTS: 1

There are work products that are delivered to the customer, e.g. a requirements document for the system.

- **Deliverables**
- Milestones
- Quality plan

ID: 22469314
PTS: 1

WSDL description The logical specification

is not converted to a WSDL description.

- True
- **False**

ID: 22469315
PTS: 1

WS-BPEL is an XML-standard for workflow specification.

- False
- **True**

ID: 22469316
PTS: 1

Service testing is difficult as (external) services are ‘white-boxes’.

- **False**
- True

ID: 22469317
PTS: 1

It may be difficult to invoke compensating actions in external services as these may rely on the failure of other services which cannot be simulated.

- False
- **True**

ID: 22469318
PTS: 1

Coordination Services that support composite processes such as ordering.

- False
- **True**

ID: 22469319
PTS: 1

The most commonly used representation for project schedules.

- graphical Notation
- **bar chart**
- validation plan

ID: 22469320
PTS: 1

A company takes on a project in a new domain. The client has not defined the process to be used and has not allowed time for risk analysis. The company has a CMM level 3 rating.

- **False**
- True

ID: 22469321
PTS: 0

Project attributes Concerned with the particular characteristics of the software development project.

- False
- **True**

ID: 22469322
PTS: 1

There are points in the schedule against which you can assess progress.

- **Milestones**
- Deliverables
- Quality plan

ID: 22469324
PTS: 1

In inter-organisational systems, not separate workflows are created for each organisation and linked through message exchange.

- True
- **False**

ID: 22469325
PTS: 1

The non-functional behavior of the service is unpredictable because it depends on load.

- False
- **True**

ID: 22469326
PTS: 0

The time required is not independent of the number of people working on the project.

- False
- **True**

ID: 22469334
PTS: 1

The logical specification is converted to a WSDL description

- **WSDL description**
- Message design
- Logical interface design

ID: 22469335
PTS: 1

Testing is intended to find defects and demonstrate that a system meets its functional and non-functional requirements.

- **True**
- False

ID: 22469336
PTS: 1

External services may be modified by the service provider thus invalidating tests which have been completed.

- False
- **True**

ID: 22469337
PTS: 1

Used once the system architecture has been designed and more information about the system is available.

- **Post-architecture model**
- Early design model
- Reuse model

ID: 22469338
PTS: 1

Estimates are made to discover the cost, to the developer, of producing a software system.

- project startup
- **Software Pricing**
- Project Planning

ID: 22469339
PTS: 0

ESLOC = ASLOC * (1-AT/100) * AAP.

- False
- **True**

ID: 22469340
PTS: 1

Describes the configuration management procedures and structures to be used.

- **Configuration management**

plan

- Quality plan
- Validation plan

ID: 22469486
PTS: 0

The development cost of usable components may be higher than the cost of specific equivalents.

- **True**
- False

ID: 22469487
PTS: 1

This avoids problems with scalability and performance if a thin-client two-tier model is chosen, or problems of system management if a fat-client model is used.

- **True**
- False

ID: 22469489
PTS: 1

The major problem with component validation is that the component specification may not be sufficiently detailed to allow you to develop a complete set of component tests.

- False
- **True**

ID: 22469490
PTS: 1

It is where the presentation layer is implemented on the client and all other layers (data management, application processing and database) are implemented on a server.

- Failure management
- **Thin-client model**
- Fat-client model

ID: 22469491
PTS: 1

Graphical workflow notations, such as BPMN, are more readable and WS-BPEL can be generated from them.

- False
- **True**

ID: 22469493
PTS: 1

PM = (ASLOC * AT/100)/ATPROD.

- False
- **True**

ID: 22469494
PTS: 0

ASLOC is the number of lines of generated code.

- True
- **False**

ID: 22469534
PTS: 1

In 1996, the 1st test flight of the Ariane 5 rocket ended in disaster when the launcher went out of control 37 seconds after take off.

- **True**
- False

ID: 22469540
PTS: 0

Access control, where service customers create individual accounts for their staff and define the resources and functions that are not accessible to each of their users.

- **True**
- False

ID: 22469541
PTS: 1

Personnel attributes Multipliers that take the experience and capabilities of the people working on the project into account.

- False
- **True**

ID: 22469542
PTS: 1

ET is the percentage of code automatically generated.

- True
- **False**

ID: 22469543
PTS: 1

ETPROD is the productivity of engineers in integrating this code.

- **False**
- True

ID: 22469563
PTS: 1

In this approach, a formulaic approach is used to compute the project effort based on estimates of product attributes, such as size, and process characteristics, such as experience of staff involved.

- Story-based planning
- Experience-based techniques
- **Algorithmic cost modeling**

ID: 22469564
PTS: 1

Do not estimate of equivalent number of lines of new code computed using the reuse model.

- **False**
- True

ID: 22469642
PTS: 1

Service testing is difficult as (external) services are ‘black-boxes’.

- **True**
- False

ID: 22469675
PTS: 1

Branding, where users from each organization, are presented with an interface that reflects their own organization.

- **True**
- False

ID: 22469721
PTS: 1

This services may be entity- or task-oriented, coordination services are always task-oriented.

- **Utility or business Services**
- Entity-oriented Services
- Task-oriented Services

ID: 22469737
PTS: 1

Estimate of equivalent number of lines of

new code computed using the reuse model.

- **True**
- False

ID: 22533202
PTS: 1

Used when requirements are available but design has not yet started

- Application composition model
- Reuse model
- **Early design model**